Woo Seok Jung | Curriculum Vitae

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EDUCATION

Sogang University **Seoul, South Korea**
M.S. in Chemistry *Mar 2025 – Present*
– GPA: 4.17/4.30
– Advisor: Prof. Bong June Sung

Sogang University **Seoul, South Korea**
B.S. in Chemistry *Mar 2021 – Feb 2025*
– GPA: 3.36/4.30 (Chemistry: 3.57/4.30)
– Micro-major, Artificial Intelligence Convergence
– Undergraduate Research Assistant, Soft Matter Computational Chemistry Laboratory

RESEARCH INTERESTS

Microscopic Mechanisms Underlying Macroscopic Properties and Physical Phenomena
Understanding how microscopic structure and dynamics govern macroscopic phenomena in complex materials. Particularly interested in phase behavior in membranes and polymers, ion transport in solid electrolytes, and mechanical behavior in polymers and metallic glasses.

Statistical Mechanics and Molecular Simulation Method Development
Developing and applying diverse computational methods, including molecular simulation and machine learning, to connect microscopic behavior with macroscopic properties.

PUBLICATIONS & MANUSCRIPTS

- **W. Jung** and B. J. Sung*, Time to Revisit: Do We Really Equilibrate a Vesicle? (manuscript in preparation).
- J. Yang†, **W. Jung**†, E. S. Song, S. Kwon, Y. Oh, and B. J. Sung*, Curvature Decouples Phase Behavior and Cholesterol Flip-Flop between the Inner and Outer Leaflets of Small Unilamellar Vesicles (submitted).

AWARDS

- **Outstanding Presentation Award, Division of Physical Chemistry, KCS** *Jun 2026*

CONFERENCE PRESENTATIONS

- **W. Jung** and B. J. Sung, Development of a Novel Vesicle Equilibration Method Using Hybrid Monte Carlo-Molecular Dynamics Simulation, KCS 138th General Meeting & Exhibition. **Poster** *Oct 2026*
- **W. Jung** and B. J. Sung, Do We Really Equilibrate Vesicles?, A3 2026 Summer School at Harbin Institute of Technology (HIT). **Oral** *Jul 2026*
- **W. Jung** and B. J. Sung, Development of a Novel Vesicle Equilibration Method Using Hybrid Monte Carlo-Molecular Dynamics Simulation, 149th Summer Symposium of the Division of Physical Chemistry, Korean Chemical Society. **Oral** *Jun 2026*
- **W. Jung** and B. J. Sung, Novel Equilibration Method for Ternary Vesicle Simulations Using Hybrid Monte Carlo-Molecular Dynamics, Annual Spring Meeting of the Polymer Society of Korea. **Poster** *Apr 2026*
- **W. Jung** and B. J. Sung, System-Dependent Challenges in Equilibration of Vesicle Simulation: DPPC-DIPC Studies Using the Martini Force Field, KCS 136th General Meeting & Exhibition. **Poster** *Oct 2025*

RESEARCH PROJECT

Identification of Microscopic Fatigue Mechanisms in Metallic Glasses Using Machine-Learned Interatomic Potentials (MLIP) *Mar 2026 – Present*

- Developed a DFT-trained Moment Tensor Potential (MTP) for Zr–Ni–Al metallic glasses across a wide range of compositions, validating structural and dynamical properties.
- Currently investigating microscopic fatigue mechanisms under cyclic loading by analyzing atomic scale correlations between local structure and non-affine deformation.

Development of a Hybrid Monte Carlo–Molecular Dynamics (MC–MD) Method for Reliable Lipid Vesicle Equilibration *Mar 2025 – Present*

- Developed a hybrid MC–MD method using interleaflet lipid swaps for efficient equilibration of ternary lipid vesicles.
- Evaluated traditional pore equilibration method by analyzing the convergence of leaflet composition, lipid numbers, and internal water content, and validated equilibrium properties against hybrid MC–MD.

Curvature-Induced Leaflet Asymmetry in Flip-Flop and Phase Behavior of Lipid Vesicles *Jul 2024 – Present*

- Identified curvature-induced leaflet asymmetry in cholesterol flip-flop dynamics and phase behavior using coarse-grained MD simulations.
- Analyzed leaflet-specific structural properties (APM, P2), cholesterol flip-flop dynamics and free-energy profiles across varying cholesterol concentrations.

Artificial Intelligence (AI) Applications in Chemistry: Trends, Opportunities, and Challenges (*Sponsored by Rebellions Inc., NPU Developer*) *Jun 2025 – Dec 2025*

- Conducted an independent analysis of AI applications and emerging technologies across major fields of chemistry.
- Evaluated representative case studies, scalability, potential applications, and limitations of AI-based approaches.

TEACHING EXPERIENCE

Sogang University **Seoul, South Korea**
Teaching Assistant, Scientific Writing in English *Mar 2026 – Aug 2026*

- Guided 40 students in improving the organization, clarity, and academic style of their scientific writing in English.

Sogang University **Seoul, South Korea**
Teaching Assistant, General Chemistry Laboratory *Sep 2025 – Feb 2026*

- Assisted 16 undergraduated students with general chemistry experiments and laboratory techniques.

ADDITIONAL EXPERIENCE

VERADI Co., Ltd. **Seoul, South Korea**
Chemistry Assessment Development Team Lead *Dec 2020 – Mar 2025*

- Developed chemistry assessment items for Korea's College Scholastic Ability Test (CSAT) and coordinated the production of published mock exams, including the Helios series.

Republic of Korea Air Force **South Korea**
Air Operations Support (CQ) *Aug 2018 – Jun 2020*

- Communicated flight schedules and operational alerts to flying squadrons.

TECHNICAL SKILLS & LANGUAGES

Programming	LaTex, Fortran, Python
Languages	Korean (Native), English
Softwares	Pytorch, LAMMPS, GROMACS, OpenMM, OVITO, VMD